

An Introduction to Climate Modeling with Julia

Milestone 5 - Constructing a System Matrix with FD Approach

1 Constructing a System Matrix with FD Approach

In this milestone, you will build the heart of the climate model: the system matrix for solving the energy balance model. The EBM model reads

$$C(x) \frac{\partial T}{\partial t} + A(CO_2) + BT - \underbrace{\vec{\nabla} \cdot (D(x) \nabla T)}_L = S_{\text{sol}}(x, t) \quad (1)$$

in spherical coordinates on the sphere's surface with the pair $(\tilde{\theta}, \varphi)$

$$L(\tilde{\theta}, \varphi) = \csc^2(\tilde{\theta}) \frac{\partial}{\partial \varphi} \left(\tilde{D} \frac{\partial T}{\partial \varphi} \right) + \frac{\partial}{\partial \tilde{\theta}} \left(\tilde{D} \frac{\partial T}{\partial \tilde{\theta}} \right) + \cot(\tilde{\theta}) \tilde{D} \frac{\partial T}{\partial \tilde{\theta}} \quad (2)$$

When we apply the finite difference method, we have for each degree of freedom:

$$\frac{\partial T_{j,i}}{\partial t} \approx \underbrace{\frac{L_{j,i}}{C_{j,i}} - \frac{B}{C_{j,i}} T_{j,i}}_{R_{j,i}(T)} + \underbrace{\frac{1}{C_{j,i}} (S_{\text{sol},j,i}(t) - A)}_{F_{j,i}} \quad (3)$$

The discretization of the heat conduction term on the surface of a sphere with a Cartesian mesh of cell size h (uniform for latitude and longitude) reads:

- For the inner degrees of freedom:

$$\begin{aligned} L_{j,i} = & \frac{\csc^2(\tilde{\theta}_{j,i})}{h^2} \left(-2\tilde{D}_{j,i} T_{j,i} + \left[\tilde{D}_{j,i} - \frac{1}{4}(\tilde{D}_{j,i+1} - \tilde{D}_{j,i-1}) \right] T_{j,i-1} \right. \\ & \left. + \left[\tilde{D}_{j,i} + \frac{1}{4}(\tilde{D}_{j,i+1} - \tilde{D}_{j,i-1}) \right] T_{j,i+1} \right) \\ & + \frac{1}{h^2} \left(-2\tilde{D}_{j,i} T_{j,i} + \left[\tilde{D}_{j,i} - \frac{1}{4}(\tilde{D}_{j+1,i} - \tilde{D}_{j-1,i}) \right] T_{j-1,i} \right. \\ & \left. + \left[\tilde{D}_{j,i} + \frac{1}{4}(\tilde{D}_{j+1,i} - \tilde{D}_{j-1,i}) \right] T_{j+1,i} \right) \\ & + \cot(\tilde{\theta}_{j,i}) \frac{\tilde{D}_{j,i}}{2h} [T_{j+1,i} - T_{j-1,i}] \end{aligned} \quad (4)$$

- For the degrees of freedom at the poles:

$$L_{1,i} = \frac{\sin(h/2)}{4\pi \text{area}[1]} \sum_{k=1}^{\text{n_longitude}} \frac{1}{2} \left(\tilde{D}_{1,k} + \tilde{D}_{2,k} \right) [T_{2,k} - T_{1,k}], \quad (5)$$

$$L_{\text{n_latitude},i} = \frac{\sin(h/2)}{4\pi \text{area}[\text{n_latitude}]} \sum_{k=1}^{\text{n_longitude}} \frac{1}{2} \left(\tilde{D}_{\text{n_latitude},k} + \tilde{D}_{\text{n_latitude}-1,k} \right) [T_{\text{n_latitude}-1,k} - T_{\text{n_latitude},k}] \quad (6)$$

for all $i \in \{1, \dots, \text{n_longitude}\}$, where $\text{area}[1] := \text{area}[\text{n_latitude}] := 0.5(1 - \cos(h/2))$ contains the normalized area of the polar cap.

To implement the model, proceed as follows:

1. Implement the mesh as a struct in Julia or a class in Python with a constructor.

Parameter	Description
n_latitude	Number of DOFs in longitude
n_longitude	Number of DOFs in latitude
ndof	Total number of DOFs
h	Cell size in latitude direction in radians
geom	Geometrical parameter at poles with $\text{geom} = \frac{\sin(h/2)}{4\pi\text{area}[1]}$
csc2	Metric term for the transformation to spherical coordinates, the vector with entries $\csc^2(\tilde{\theta}_j)$
cot	Metric term for the transformation to spherical coordinates, the vector with entries $\cot(\tilde{\theta}_j)$
area	Area of each cell on the surface of the sphere which only depends on latitude

2. Calculate the earth surface type dependent heat conduction coefficients $\tilde{D}(\tilde{\theta}, \varphi) \in \mathbb{R}^{n_y \times n_x}$ by implementing a function `calc_diffusion_coefficients` that maps the following equation:

$$\tilde{D}(\tilde{\theta}, \varphi) \in \mathbb{R}^{n_y \times n_x} = \begin{cases} \tilde{D}_{\text{ocean,poles}} + (\tilde{D}_{\text{ocean,equ}} - \tilde{D}_{\text{ocean,poles}}) \sin(\tilde{\theta})^5 & \text{for the ocean} \\ \tilde{D}_{\text{NP}} + (\tilde{D}_{\text{equ}} - \tilde{D}_{\text{NP}}) \sin(\tilde{\theta})^5 & \text{for the Northern Hemisphere} \\ \tilde{D}_{\text{SP}} + (\tilde{D}_{\text{equ}} - \tilde{D}_{\text{SP}}) \sin(\tilde{\theta})^5 & \text{for the Southern Hemisphere} \end{cases} \quad (7)$$

The corresponding thermal conductivity coefficient¹ are as follows:

Types	Thermal conductivity in $W/m^2/K$
Ocean at Poles	0.4
Ocean at Equator	0.65
Equator	0.65
North Pole	0.28
South Pole	0.2

Table 1: Surface specific thermal conductivity for the determination of diffusion coefficients

Note: For the surface type dependency of the heat conduction, you will once again need to include the function `read_geography` from milestone 1.

3. Use the above formulas, the function `calc_diffusion_coefficients` and the functions for the other parameters from previous milestones to write a function `compute_equilibrium_EBM_2D` to compute the EBM for two spatial dimensions using a forward Euler time integration scheme. Try to calculate the EBM using this function. What happens?
4. Write a function `calc_jacobian_ebm_2d` that computes the Jacobian matrix of the FD operator numerically. You can calculate the j -th column of the Jacobian matrix with the equation:

$$\underline{\mathbf{A}}_{\text{EBM}} \hat{\mathbf{K}}^j = \mathbf{R}(\hat{\mathbf{K}}^j) - \mathbf{R}(\mathbf{0})$$

where $\hat{\mathbf{K}}^j = (\hat{k}_1^j, \hat{k}_2^j, \dots, \hat{k}_{\text{NDOF}}^j)^T$ is a vector whose entries are defined as

$$\hat{k}_i^j = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

¹K. Zhuang, G.R. North, M.J. Stevens, *A NetCDF version of the two-dimensional energy balance model based on the full multigrid algorithm*, SoftwareX, Vol. 6, pp. 198-202, July 7, 2017.

Note that the matrix F does not depend on T and cancels out.

Here, we treat $\mathbf{R}$ as a function $\mathbf{R}: \mathbb{R}^{\text{NDOFS}} \rightarrow \mathbb{R}^{\text{NDOFS}}$. More precisely, in differential geometry terms, $R^{m \times n}$ is an mn -dimensional manifold, and in order to compute a Jacobian of a function on that space (or do calculus in general), we need coordinate charts mapping this space to $\mathbb{R}^{\text{NDOFS}}$. Under these charts, we can calculate the Jacobian. A simple chart for this space maps a matrix to a vector containing all its entries. The inverse of this chart then reorganizes these entries into a matrix. In code, we can do the same with the functions `flatten` and `reshape`, which are available in all languages for numerical computing.

This function requires many calls to `calc_diffusion_coefficients`. It is therefore important that the diffusion coefficients can be computed in the microsecond range. Unfortunately, Python, being an interpreted language, is slow. To bring `calc_diffusion_coefficients` into the microsecond range, we recommend using Numba's `@jit` decorator. Note that this only works when the function parameters are of certain types, mostly Numpy types. Therefore, this function cannot take the mesh as a parameter and has to take the mesh's fields directly.

5. Examine the sparsity pattern of the Jacobian matrix. What does the matrix look like? Be aware that `spy` of `Plots.jl` can be very slow for large matrices when using the `PythonPlot` backend. If you want to examine the whole Jacobian, use the `GR` backend instead.
6. Compute the eigenvalues of the Jacobian matrix. Determine a practical approximation to the largest time step for which the scheme is stable using the forward Euler time integration scheme.

2 Control Solutions

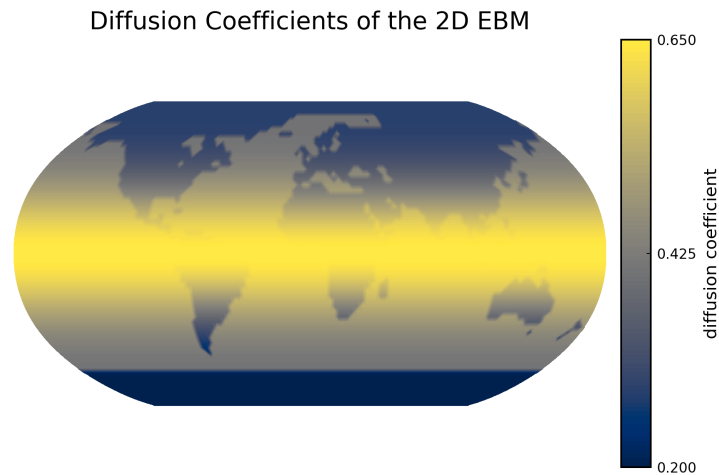


Figure 1: Diffusion coefficients

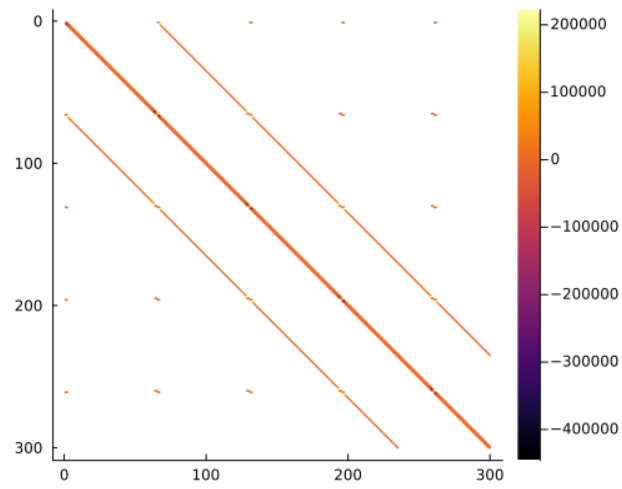


Figure 2: Sparsity pattern of the upper left part of the system matrix

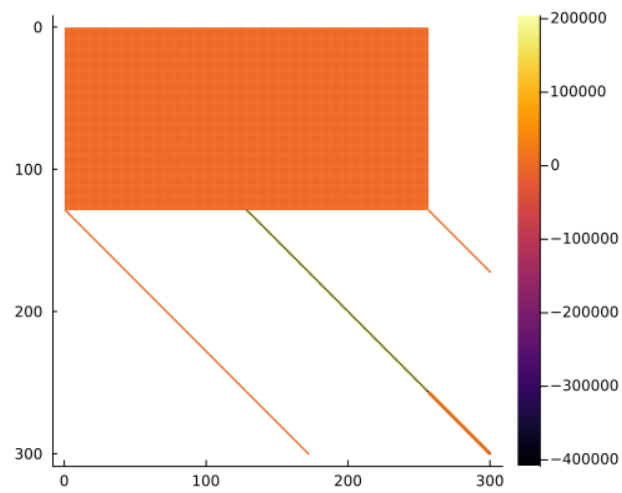


Figure 3: Sparsity pattern of the upper left part of the system matrix in a row-major language like Python